A bounded-variation primitive description for the rank-one left-shift sun-dual example

FA/Banach run `fa_banach_001`, lane 14

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Source Metadata

- Source paper: Sahiba Arora and Felix L. Schwenninger, *Admissible operators for sun-dual semigroups*, arXiv:2408.02150v3.
- Local source PDF: `source_paper.pdf`, compiled from the local arXiv v3 source.
- Extracted question location: Proposition 5.3 and Remark 5.4(c), page 14 of the compiled arXiv v3 PDF.
- Claimed status: candidate full answer to the rank-one left-shift simplification question in Remark 5.4(c).

Open Problem Extract

In the source paper's example, $X = L^1([0,1])$, $(R(t))_{t\geq 0}$ is the nilpotent right-shift semigroup, and the sun-dual semigroup is the nilpotent left-shift semigroup $(L(t))_{t\geq 0}$ on

$$X^{\odot} = \{b \in C([0,1]) : b(1) = 0\}.$$

The source describes the rank-one C-admissible controls for $(L(t))_{t\geq 0}$ as

$$\{C' : C \in \mathcal{L}(X_1, \mathbb{C}),\ C'(1) = \partial b = \widetilde{\partial} c \text{ for some } b \in X^{\odot},\ c \in \mathrm{BV}([0,1])\}.$$

Remark 5.4(c) asks whether this description can simply be written as

$$C'(1) = \partial b \quad \text{for some } b \in X^{\odot} \cap \mathrm{BV}([0,1]).$$

the direct computation of admissible control operators associated to $(L(t))_{t\geq 0}$ on $X^\odot$ is tedious, our results in the prequel along with the analysis in the present section allow us to characterise all C-admissible rank-one control operators:

Proposition 5.3. The set $\mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^\odot, (L(t))_{t\geq 0})$ can be described as

$$\{C' : C \in \mathcal{L}(X_1, \mathbb{C}) \text{ and } C'(1) = \partial b = \widetilde{\partial} c \text{ for some } b \in X^\odot, c \in \mathrm{BV}\big([0,1]\big)\}.$$

Proof. Firstly, let $C \in \mathcal{L}(X_1, Y)$ be such that $C'(1) = \partial b = \widetilde{\partial} c$ for some $b \in X^\odot$ and $c \in \mathrm{BV}\big([0,1]\big)$. By the description of $(X^\odot)_{-1}$, we get $C'(\mathbb{C}) \subseteq (X^\odot)_{-1}$. In fact, the continuity of b, in particular, also means that $\widetilde{\partial} c = \partial b$ has no atomic part. Together, Proposition 5.2 and Theorem 4.3 now imply that $C' \in \mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^\odot, (L(t))_{t\geq 0})$.

Conversely, let $B \in \mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^\odot, (L(t))_{t\geq 0})$, then sun-reflexivity of X – which is known from [58, Example 1.3(ii)] – along with Theorem 3.1 yields that $B' \in \mathbb{C}_1(X, \mathbb{C}, (R(t))_{t\geq 0})$. Thus there exists $c \in \mathrm{BV}\big([0,1]\big)$ such that $B''(1) = \widetilde{\partial} c$. Also, as $B''(1) = B(1) \in (X^\odot)_{-1}$, there exists $b \in X^\odot$ such that $B''(1) = \partial b$. Thus, $B = B''$ has the desired form. ☐

Remarks 5.4. (a) The proof of Proposition 5.3 even shows that the admissibility of each element of $\mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^\odot, (L(t))_{t\geq 0})$ is zero-class; cf. Theorem 4.3.

(b) It can be inferred from [5, Corollary 4.7] that every positive $B \in \mathcal{L}(\mathbb{C}, (X^\odot)_{-1})$ is zero-class C-admissible.

(c) While $g = \widetilde{f}$ implies $g = \partial f$, the converse is not true in general. Therefore, we do not know whether in the description of $\mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^\odot, (L(t))_{t\geq 0})$ from Proposition 5.3 we can simply write $C'(1) = \partial b$ for some $b \in X^\odot \cap \mathrm{BV}\big([0,1]\big)$.

Proof Intuition

The only apparent gap between the source description and the proposed simpler one is whether the primitive b must have bounded variation. But the source already gives a bounded-variation primitive c for the same distribution after passing through the $\widetilde{\partial}$ calculus. Restricting that equality to ordinary compactly supported test functions says that the distributional derivative of the continuous function b is a finite Radon measure on $(0,1)$. In one dimension, that is exactly the distributional characterization of bounded variation.

The converse is just integration by parts. If $b \in X^\odot \cap \mathrm{BV}([0,1])$, then ∂b and $\widetilde{\partial} b$ agree on the relevant test functions. The endpoint term at 1 is killed by $b(1) = 0$, while the endpoint term at 0 is killed by the definition of the $\widetilde{\mathcal{D}}$ test space.

Theorem

Theorem 1. *With the notation of Arora–Schwenninger, for the nilpotent left-shift semigroup $(L(t))_{t\geq 0}$ on*

$$X^\odot = \{b \in C([0,1]) : b(1) = 0\},$$

one has

$$\mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^\odot, (L(t))_{t\geq 0}) = \{C' : C \in \mathcal{L}(X_1, \mathbb{C}),\ C'(1) = \partial b \text{ for some } b \in X^\odot \cap \mathrm{BV}([0,1])\}.$$

Thus the simplification asked for in Remark 5.4(c) is valid.

Proof

Let

$$\widetilde{\mathcal{D}} = \{\phi \in C^\infty([0,1]) : \phi(0) = 0\}, \qquad \mathcal{D} = C_c^\infty((0,1)).$$

For distributions on $\widetilde{\mathcal{D}}$, the notation $g = \widetilde{\partial} f$ means

$$\langle g, \phi \rangle = -\langle f, \phi' \rangle \qquad (\phi \in \widetilde{\mathcal{D}}),$$

whereas ∂f denotes the ordinary distributional derivative on $\mathcal{D}$.

The source Proposition 5.3 proves that

$$\mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^{\odot}, (L(t))_{t\geq 0}) = \{C' : C \in \mathcal{L}(X_1, \mathbb{C}),\ C'(1) = \partial b = \widetilde{\partial} c \text{ for some } b \in X^{\odot},\ c \in \mathrm{BV}([0,1])\}. \tag{1}$$

We show that the set on the right of (1) is exactly the simpler set in the theorem.

First suppose that $C'(1) = \partial b = \widetilde{\partial} c$ as in (1). Since $c \in \mathrm{BV}([0,1])$, the distribution $\widetilde{\partial} c$, restricted to $\mathcal{D}$, is a finite Radon measure on $(0,1)$. Hence the ordinary distributional derivative of the continuous function b is a finite Radon measure on $(0,1)$.

By the standard one-dimensional characterization of functions of bounded variation, an L^1_{loc} function whose distributional derivative is a finite Radon measure is a BV function, with that measure as its distributional derivative. Applying this to b on $(0,1)$ shows that b has bounded variation on $(0,1)$. Since b is continuous on $[0,1]$, its one-sided endpoint limits are finite and agree with $b(0)$ and $b(1)$; adjoining the endpoints therefore does not add infinite variation. Thus

$$b \in X^{\odot} \cap \mathrm{BV}([0,1]).$$

This proves that every control in the source description has the simplified form.

Conversely, suppose that $b \in X^{\odot} \cap \mathrm{BV}([0,1])$ and $C'(1) = \partial b$. The BV integration-by-parts formula gives, for every $\phi \in \widetilde{\mathcal{D}}$,

$$-\int_0^1 b(t)\phi'(t)\,dt = \int_{[0,1]} \phi(t)\,dDb(t) - b(1)\phi(1) + b(0)\phi(0),$$

where Db is the finite signed distributional derivative measure of b. Now $b(1) = 0$ because $b \in X^{\odot}$, and $\phi(0) = 0$ because $\phi \in \widetilde{\mathcal{D}}$. Hence

$$\langle \widetilde{\partial} b, \phi \rangle = \int_{[0,1]} \phi(t)\,dDb(t).$$

On compactly supported test functions this is exactly ∂b. Therefore $\partial b = \widetilde{\partial} b$ in the sense needed in Proposition 5.3. Taking $c = b \in \mathrm{BV}([0,1])$ in (1), the source proposition gives

$$C' \in \mathbb{B}_{\mathrm{C}}(\mathbb{C}, X^{\odot}, (L(t))_{t\geq 0}).$$

The two inclusions prove the asserted equality and answer the question in Remark 5.4(c). □

Verification Notes

- No numerical or symbolic computation is used.
- The proof depends only on the source Proposition 5.3 and the standard distributional characterization of one-dimensional BV functions.
- The boundary terms are the main point: $b(1) = 0$ is part of the definition of $X^{\odot}$, and $\phi(0) = 0$ is part of the definition of $\widetilde{\mathcal{D}}$.
- This packet answers only the rank-one left-shift description in Remark 5.4(c). It does not settle the broader $p = 1$ zero-class-removal problem from Theorem 4.3.

Novelty Check

Local run indexes were searched for `2408.02150`, the source title, “admissible operators”, “sun-dual”, “zero-class”, “BV”, “partial b”, and related semigroup terms. No duplicate packet or attempt for this exact remark was found.

Bounded web searches on 2026-07-19 for the exact title, `C'(1)=partial b`, “sun-dual semigroups zero-class assumption p=1”, and close variants found the arXiv v3 source and publication metadata for the journal version, but no later paper or erratum resolving Remark 5.4(c). The arXiv record lists version 3 as the current version and gives the journal reference in *Mathematics of Control, Signals, and Systems* (2025).

Human Review Recommendation

Review as a likely full solution of the narrow Remark 5.4(c) question. The main item to check is that the equality $\partial b = \widetilde{\partial} c$ in Proposition 5.3 is interpreted with the standard restriction to ordinary test functions on $(0, 1)$; under that interpretation the BV characterization proves the converse exactly.

References

[1] Sahiba Arora and Felix L. Schwenninger, *Admissible operators for sun-dual semigroups*, arXiv:2408.02150v3, 2025.

[2] Luigi Ambrosio, Nicola Fusco, and Diego Pallara, *Functions of bounded variation and free discontinuity problems*, Oxford Mathematical Monographs, Oxford University Press, 2000.